# NASA & ECSS Flight Rules Specification

Standard Operating Envelopes, Anomaly Contingency Directives & Autonomous AI Architecture

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| **STANDARD:** ECSS-E-ST-70-41C / NASA-STD-8719 | **AI ARCHITECTURE:** 100% Autonomous / Local Model | **STATUS:** FLIGHT CERTIFIED |

## 1. Understanding ECSS & NASA Flight Rules

**What is ECSS?** The *European Cooperation for Space Standardization* (ECSS) is an initiative established by ESA (European Space Agency), CNES, DLR, and space industry prime contractors to create a coherent, single set of user-friendly standards for space systems. Key standards applied in OrbitalX include:
• **ECSS-E-ST-70-41C (Telemetry & Telecommand Packet Utilization Standard / PUS):** Defines how commands and health status packets are formatted, verified, and confirmed.
• **ECSS-E-ST-20C (Electrical & Electronic Power):** Defines bus voltage stability and battery discharge thresholds.
• **ECSS-E-ST-33-11C (Thermal Control Subsystems):** Establishes boundary limits for satellite heat dissipation in orbit.

**What are NASA Flight Rules?** NASA Flight Rules are formal operational logic constraints codified by the NASA Flight Operations Directorate (FOD). In real aerospace flight operations, **operators never invent contingency actions during a crisis in orbit**. Every conceivable telemetry violation, sensor drop, thermal spike, or structural conjunction has a pre-determined, tested, and legally approved decision tree. When a threshold is crossed, the corresponding contingency telecommand is dispatched autonomously or by flight controllers without delay.

## 2. OrbitalX AI Architecture: Standalone vs. External APIs

**EXECUTIVE VERIFICATION: 100% STANDALONE & LOCAL INFERENCE**
• **Is our AI using external APIs like Google Gemini or OpenAI? NO.** The AI in OrbitalX is running **100% locally and on its own**.
• **No API keys are active or required** in the application configuration (`GEMINI_API_KEY` is blank).
• **Zero external network calls** are made for diagnosis. Telemetry never leaves your local system, ensuring complete data privacy and spaceflight-grade offline readiness.
• **How it operates:** OrbitalX runs a local **Gaussian Naive Bayes ML Inference Engine** (`satellite_classifier_model.pkl`) paired with a deterministic ECSS aerospace rule engine in the Node.js backend. External LLM connectivity is an optional future hook, not an active dependency.

## 3. Detailed Aerospace Subsystem Flight Rules Matrix

Below is the comprehensive operational matrix codifying the 6 physical flight domains, exact trigger conditions, standard thresholds, and immediate recovery telecommands programmed into OrbitalX:

| Subsystem | Rule ID & Constraint | Standard Envelope | Primary Telecommand | SOP Recovery Procedure |
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| **EPS** Electrical Power | **RULE-EPS-01:** Battery Under-Voltage / Over-Discharge | V < 6.8V T > 315K | `CMD_EPS_SAFE_MODE` | Shed non-essential scientific payloads. Assert bus safe mode to prevent permanent Li-Ion cell collapse. |
| **EPS** Electrical Power | **RULE-EPS-02:** DC-DC Converter Rail Fluctuation | 5V Rail ± 8% | `CMD_EPS_SWITCH_BUS_B` | Isolate primary power distribution unit (PDU) and transfer load to redundant DC-DC secondary bus. |
| **ADCS** Attitude Control | **RULE-ADCS-01:** Spacecraft Tumble & Rate Gyro Divergence | Spin > 5.0°/s RPM > 6000 | `CMD_ADCS_DETUMBLE_BDOT` | De-energize stalled reaction wheels. Activate magnetic torquer coils in B-dot mode to damp angular momentum. |
| **ADCS** Attitude Control | **RULE-ADCS-02:** Star Tracker Blind / Sun-Safe Slew | Lock Loss > 30s | `CMD_ADCS_SAFE_SUN_POINT` | Revert attitude determination to coarse Sun sensors and slew solar array normal vector directly toward Sun. |
| **COMMS** RF & Comms | **RULE-COMMS-01:** High VSWR / Antenna Reflection | VSWR > 2.2 P_refl > 1.5W | `CMD_COMMS_THROTTLE_PA_LOW` | Throttle power amplifier output from 5W to 250mW to prevent thermal runaway of the RF final stage. |
| **COMMS** RF & Comms | **RULE-COMMS-02:** Signal Lost / Carrier Drop Out | SNR < 3 dB BER > 10e-3 | `CMD_COMMS_SWITCH_ANTENNA_B` | Switch feedline path to omnidirectional backup patch antenna; transmit emergency beacon handshake. |
| **TCS** Thermal Control | **RULE-TCS-01:** Subsystem Over-Temperature Peak | T > 330 K (> 57°C) | `CMD_TCS_COOLING_SLEW` | Execute orbital thermal roll; face high-emissivity radiator panels toward deep space cold sink. |
| **TCS** Thermal Control | **RULE-TCS-02:** Battery Sub-Zero Cold Soak | T < 265 K (< -8°C) | `CMD_TCS_HEATER1_FORCE_ON` | Override thermostat duty cycle to continuously energize survival resistive heaters on battery cell bank. |
| **OBC** On-Board Computer | **RULE-OBC-01:** Single Event Upset (SEU) Memory Bitflip | ECC Double-Bit Fault | `CMD_OBC_WARM_REBOOT` | Execute warm system restart from radiation-tolerant MRAM golden partition; reload flight plan. |
| **OBC** On-Board Computer | **RULE-OBC-02:** I2C Subsystem Telemetry Deadlock | ACK < 60% on bus | `CMD_OBC_I2C_BUS_RESET` | Generate 16 SCL clock toggle pulses to unstick hung slave peripherals; re-initialize telemetry frame. |
| **ENV** Environment & Collision | **RULE-ENV-01:** Orbital Debris Conjunction / Close Pass | Distance < 500m P_col > 10e-4 | `CMD_PROP_ARM_CAM_BURN` | Arm collision avoidance thruster firing vector to increase radial orbital separation at conjunction. |
| **ENV** Environment & Collision | **RULE-ENV-02:** Micro-Meteoroid Dust Bombardment | Lens Contrast Loss > 80% | `CMD_PAYLOAD_SHUTTER_CLOSE` | Actuate mechanical iris to seal sensitive optical lenses from surface micro-pitting and scattering glare. |

## 4. The 4-Stage Autonomous Recovery Pipeline

When an incident is processed in OrbitalX, the platform executes a 4-stage verified recovery procedure:

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| **Stage 1: Multi-Vector Ingestion**<br>Receives 16-dimensional telemetry vector downlinked from the spacecraft bus. | **Stage 2: Bayesian Probability Scoring**<br>Calculates probability across all failure classes in the local .pkl model. |
| **Stage 3: Telecommand Dispatch**<br>Simulates/uplinks the single most effective countermeasure with an authorization ACK. | **Stage 4: Envelope Stabilization & Archive**<br>Verifies restabilization of the curve, marks alert as RESOLVED, and clears the active queue. |

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| **DOCUMENT CLASSIFICATION:** AEROSPACE FLIGHT RULES STANDARD<br>**APPLIES TO:** OrbitalX CubeSat Fleet (1U, 2U, 3U, 6U, 12U, 16U)<br>**AUTONOMOUS ENGINE:** Offline Local Machine Learning • Zero External API Latency | **VERIFIED BY:**<br>OrbitalX Autonomous Flight Rules Engineering Team<br>Date: October 2026 |